Key Transparency
(When Collusion-Based Security Fails)

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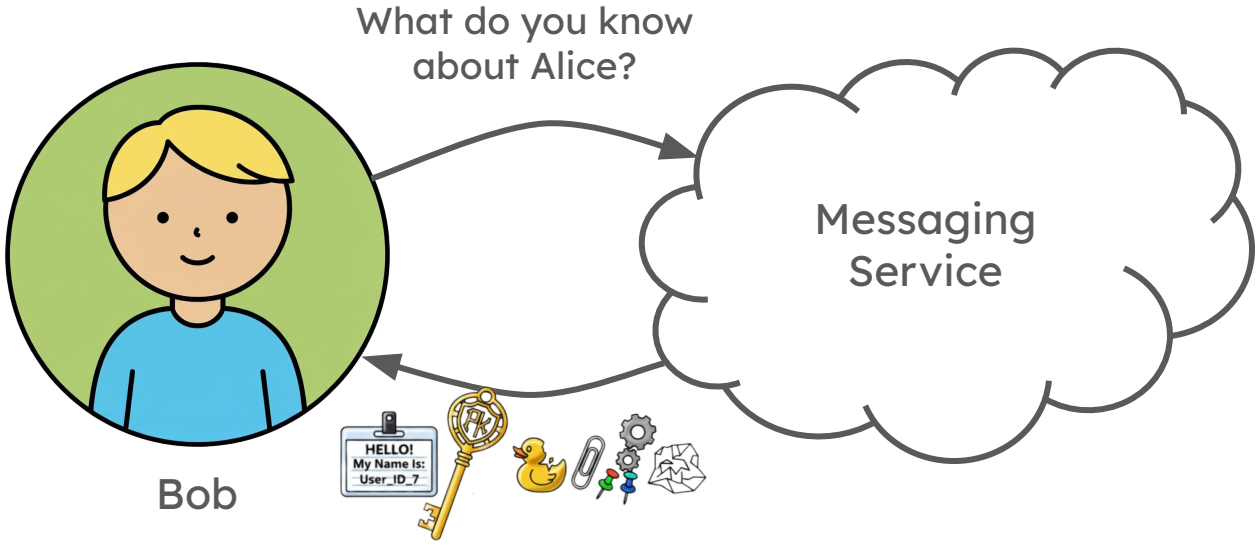
Security Model



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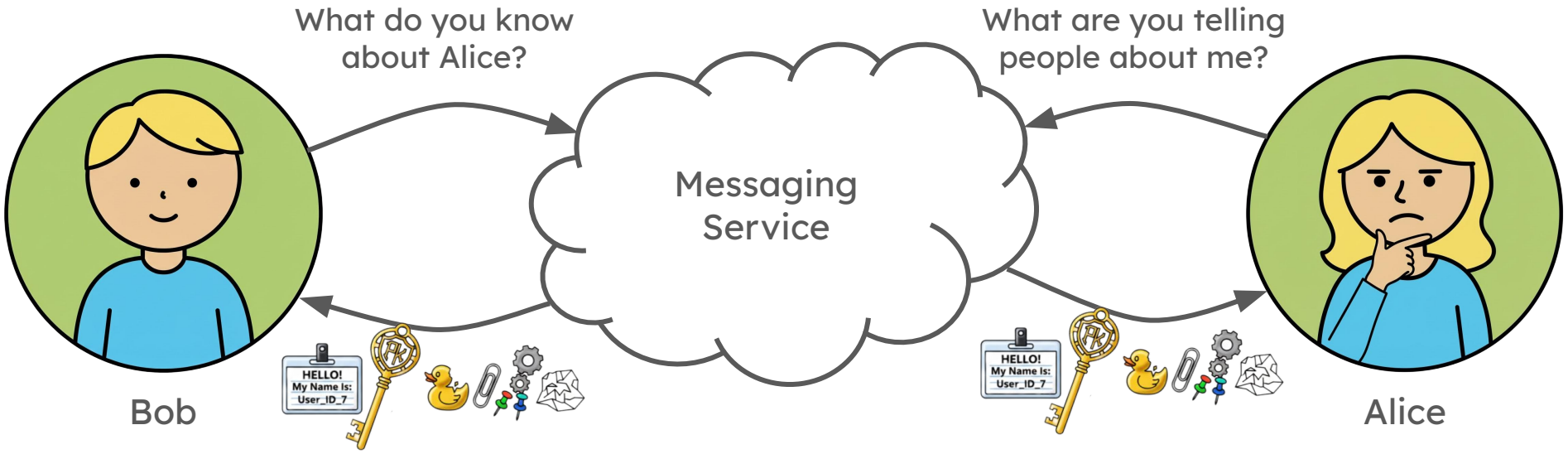


How does Alice really *know* what people are being told about her?

- Alice puts her information into some kind of Transparency Log
- Most Transparency Logs are *just logs*

  => For Alice to find everything about her, she'd need the *whole* thing

Logs need structure

But how do we know they're organized correctly?

Option 1: Let anyone check any time?

Main problem: Bandwidth cost

- Letting any stranger download a huge amount of data is expensive!

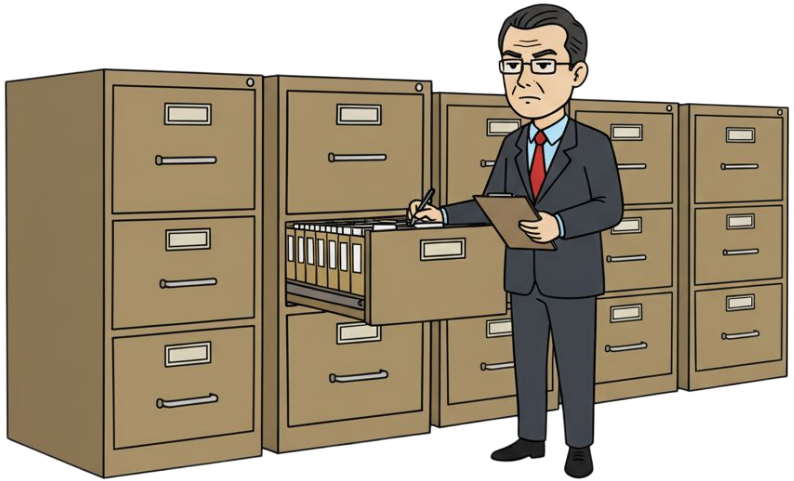
Secondary problem: Security

- Users have no reason to believe that anyone ever actually did the work :(

Option 2: Have a trusted party check?

Main problem: Operational burden

- Things service operators worry about:
  - How do I even find this person?
  - What if they have some huge outage?
  - What if they just disappear one day?
  - What if I want to change the protocol I'm using?
  - **Why would my users trust this person?**



Option 3: Have users themselves check!

Users check that the *specific* information they receive is correctly sorted and persists over time.

- Makes more work for Bob but results in a protocol that is functional and secure ✨ out of the box ✨

Step 1: Bob looks up something about Alice

Step 1½: Alice checks that everything (about her) is correct

Step 2: Bob checks that what he saw is still there

Option 3: Have users themselves check!

Users check that the *specific* information they receive is correctly sorted and persists over time.

- Makes more work for Bob but results in a protocol that is functional and secure ✨ out of the box ✨

Step 0: Alice checks that everything (about her) is correct

Step 1: Bob looks up something about Alice (and sees that it's old)

Step 2: Bob checks that what he saw is still there

Communicating pairs of users are their own trusted third party!

This model works so well for KT because:

Users are consistently online

Users have very robust and secure local state

Relatively low number of contacts

Thanks :)

Read the drafts:
https://datatracker.ietf.org/doc/draft-ietf-keytrans-architecture/
https://datatracker.ietf.org/doc/draft-ietf-keytrans-protocol/

Read the code:
https://github.com/Bren2010/katie (in-progress)